

Mr. Rofhiwa Nemukula - Specialist Data Scientist & Advanced Analytics | AI Scientist

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Professional Experience

Specialist Advanced Analytics - CellC Limited

November 2024 - Present

- Automated Power BI and Microsoft Excel daily reports by scheduling Snowflake SQL scripts to create tables using Git and Databricks.
- Developed model features (usage, activity, spend, demographics) in PySpark using Databricks for a recommender system.
- Developed a Machine Learning Operations (MLOps) framework for implementing a recommender system.
- Utilized Snowflake to create data models and perform data analysis.
- Conducted statistical analysis on customer behavior to understand interaction patterns with our products.
- Presented analysis and project updates to both technical and business audiences.
- Assisted in developing business rules for various processes.
- Implemented technical summer campaigns for cross-selling.
- Developed a Generative Artificial Intelligence model for creating SMS campaigns for different customer segments.
- Created a Customer Satisfaction Score (CSAT) analysis tool for capturing and reporting customer feedback.

Specialist Data Scientist - Multichoice Group

December 2023 - October 2024

HR BOT - AI Chatbot

- Responsible for the requirements gathering from the stakeholder, this includes the creation of stage plan and running the sign off meeting with the stakeholders. This includes the mapping of the customer journey and outlining successful metrics measures of the project.
- Created a wireframes and mockups of the final mobile and desktop application based on the customer requirements using Figma. The purpose of this task was to show the UI & UX design to the stakeholder before the implementation of the product.
- The HR Bot POC was implement using local GraphRAG.
- Used Azure Open AI GPT4 for LLM generation.
- Developed a back-end project map following the C4 model software architecture using Miro.
- Developed a system that reads documents and convert the documents to a markdown using marker framework.
- Developed an API framework.
- Developed a front-end using next.js, node js, tailwindcss, javascript and typescript. Developed on VSCODE
- Used Azure App services, and Docker for deployment.
- Used Git and Azure DevOps for Version-control.

Project Lead on AI Chatbot Development (TV Guide Bot):

- Designed and implemented a Retrieval-Augmented Generation (RAG) system to enhance the capabilities of an AI chatbot.
- Integrated external knowledge sources and databases to improve the chatbot's response accuracy and relevance. This was achieved by converting data into embeddings using a pre-trained Generalized Text Embeddings (GTE) model from Hugging Face. Embeddings generated from the text data were stored in a vector database using Facebook AI Similarity Search (FAISS).
- Implemented semantic analysis and routing to identify the intent, context, and meaning behind a query, allowing for prompt rejection before cosine similarity search is done on the vector database
- Applied re-ranking strategies on the cosine similarity search results to refine RAG model outputs, enhancing recommendation relevance and quality this was done using RAGtouille - ColBERT model.
- Utilized machine learning techniques to improve re-ranking based on historical data.
- Integrated Azure Open AI services into RAG solutions for advanced AI capabilities and improved performance.

Fine-Tuning Pre-Trained Models:

- Conducted large-scale data analysis using PySpark for efficient big data processing.
- Fine-tuned a BERT model from Hugging Face for multi-class text classification tasks.
- Conducted extensive experimentation to optimize hyper-parameters and improve model performance.
- Implemented advanced techniques such as data augmentation and ensemble learning to enhance classification accuracy.

Performance Evaluation and Improvement:

- Evaluated model performance using metrics such as accuracy, F1 score, precision, and recall.
- Continuously monitored deployed models and implemented improvements based on user feedback and performance metrics.

Technical Documentation and Presentations:

- Created comprehensive technical documentation for AI models and systems.
- Presented project progress and findings to stakeholders and senior management.

AI Research and Prototyping:

- Conducted research on the latest advancements in AI and NLP to identify potential improvements for ongoing projects.
- Built prototypes to demonstrate the feasibility and effectiveness of new AI techniques.

Data Pipeline Development:

- Designed and implemented data pipelines for efficient processing and preparation of large-scale text data using Azure Data Factory and SQL Server Integration Services
- Developed SQL scripts to source data from Microsoft SQL Management Studio.
- Utilized tools such as Apache Spark for real-time data streaming and processing.

Cross-Functional Team Collaboration:

- Collaborated with product development teams to integrate AI solutions into existing products and services.
- Worked with data engineers to ensure seamless data flow

Graduate Machine Learning Engineer - Multichoice Group

March 2023 - December 2023

- Designing dynamic ML models and systems which have the ability to train, retrain themselves whenever required on Azure Databricks, primarily using Python Programming Language.
- Implemented and maintained data pipelines in Azure Data Factory to source and transform the data required for Machine Learning models.
- Building pipelines on SQL Server Integration Services (SSIS) to source data used in Machine Learning projects
- Developing Dashboards in Microsoft Power BI as per stakeholder requirements for reporting.
- Performing Data exploration and feature engineering.
- Process optimization for improving customer experience.
- Performing data wrangling using SQL on Microsoft SQL Server Management Studio (SSMS).
- Developing the best-in-class statistical models and algorithms.
- Selecting appropriate datasets and data representation methods.
- Visualizing data for insights and to perform statistical analysis, applying new insights to improve ML model.
- Conducting research on machine learning algorithms and libraries.
- Running machine learning tests and experiments on Azure Databricks.
- Performing model optimization to existing machine learning model
- Using SQL for querying and managing relational databases to extract and process data efficiently
- Writing SQL queries for ETL (Extract, Transform, Load), data warehousing, and Business Intelligence tasks.
- Designing and developing scalable ETL packages from the business source systems.
- Working with data engineers and systems architects to monitor deployments and maintain models and systems.
- Presenting analysis and project updates to both technical and business audiences.
- Integrating Machine Learning Models into production systems using Microsoft Azure systems

Engineering Mathematics 1A Part-time Lecturer - University of Johannesburg

February 2023 - July 2023

- Developed personalized learning strategies for students struggling with specific topics, ensuring tailored support for diverse learning needs.
- Facilitated group study sessions to encourage peer learning and collaboration among students.
- Integrated digital tools and resources such as interactive simulations or online math platforms to enhance learning engagement.
- Provided real-world engineering examples to help students understand the practical applications of mathematical concepts.
- Offered one-on-one consultations for students needing extra help, fostering a deeper understanding of complex material.
- Monitored student progress using data-driven approaches, providing detailed feedback to students on their performance.
- Created a supportive learning environment that encouraged open communication, ensuring students felt comfortable seeking help.
- Coordinated with other faculty members to align extra classes and tutorials with broader curriculum goals.
- Designed engaging problem-solving activities to help students apply theoretical concepts to practical engineering problems.
- Encouraged students to explore additional resources beyond the curriculum, such as industry articles or advanced mathematical challenges.
- Conducted extra classes and tutorials based on the study guide provided by the department of Mathematics
- Prepared students for class tests, semester tests, and examinations by hosting online classes on Saturdays to
- Assisted students with time management, study plans, mentorship and career guidance to motivate students

Advanced Manufacturing 4A Teaching Assistant - University of Johannesburg

February 2022 - July 2022

SolidWorks Design and Model Generation:

- Use SolidWorks software to create detailed, functional 3D component models required for student projects and experiments.
- Provide one-on-one and group support to help students develop their design skills using SolidWorks.

Hosting 3D Printing Practicals:

- Organize and lead practical sessions where students learn to operate 3D printers.
- Explain the step-by-step process of setting up 3D printers, selecting appropriate materials, and executing print jobs.

Tensile Testing of 3D Printed Components:

- Conduct tensile strength tests on the 3D printed components, ensuring they meet industry and academic standards.
- Instruct students on the process of tensile testing, including setup, execution, and data interpretation.
- Facilitate discussions on the relationship between materials, design, and strength in 3D printed objects.

Explanations on 3D Printing and Industry Applications:

- Provide comprehensive explanations on the principles and mechanics of 3D printing technology.
- Illustrate the benefits of 3D printing in the manufacturing industry, emphasizing rapid prototyping, cost-efficiency, and innovation.

Technical Report Writing for Engineers:

- Host lectures that teach students the essentials of writing detailed technical reports, including documenting experimental methods, results, and analysis.
- Review and provide feedback on the technical content, structure, and clarity of student reports.
- Highlight the importance of precision and detail in professional engineering documentation.

Marking and Reviewing Practical Reports:

- Assess and grade the practical reports produced by students, ensuring they reflect a clear understanding of the experiments conducted.
- Offer constructive feedback to improve students' technical writing and reporting skills.

Tutorials on Manufacturing Calculations:

- Conduct extra tutorial classes on manufacturing-related calculations, focusing on cost analysis, production planning, and resource allocation.
- Teach students how to apply mathematical models and formulas in real-world manufacturing scenarios.
- Explain how manufacturing companies use these calculations to optimize production and reduce costs.

Lab and Equipment Management:

- Ensure that the 3D printing lab and its equipment are well-maintained and ready for practical use.
- Oversee the safety protocols during practicals, ensuring a safe and productive learning environment.

Projects

Claimpal AI Chatbot:

- Developed an AI-powered chatbot designed to help users better understand insurance documents. The Claimpal web app allows users to upload insurance documents from any device and engage in a conversational interface to clarify the content.
- The application processes uploaded documents, converts them into a markdown format, and makes the information more accessible.
- Firebase was used as the backend cloud storage solution to securely store and manage user data and documents.
- Postman was utilized for testing and hosting the API endpoints, ensuring smooth communication between the frontend and backend components.
- The AI model for natural language processing and generation was powered by OpenAI's GPT-4 mini, providing context-aware responses to user queries about the insurance documents.
- Both front-end and back-end components were developed using TypeScript for robust, typed code. The frontend was built with React and Next.js, enabling a dynamic, responsive user experience. For styling, Tailwind CSS was used to create a modern, efficient design system. The backend was built with Node.js to handle requests and data processing.

AI Chatbot with RAG (TV Guide Bot):

- Developed an AI chatbot utilizing RAG to provide accurate and contextually relevant responses.
- Integrated with various data sources to enhance knowledge retrieval and user interaction.
- Used PostgreSQL to store vector embeddings
- Developed a front-end user interface using Streamlit(Customized using HTML & CSS).
- Deployed the application using Docker.

Multi-Class Text Classification:

- Fine-tuned a pre-trained Hugging Face model (RoBERTa) to classify customer review into predefined categories with 83% accuracy.
- Created a PowerBI report to show the customer review mappings to the predefined categories from the model. Created a gateway to make the PowerBI refresh

Time-series Forecasting:

- Developed a multi-stage forecasting model for the Multichoice Walk in Centers (WIC) volumes for different time intervals (30 minutes, hourly, daily, monthly, & long-term)

- Reduced the forecasting error from 25% to 12%
- Automated the forecast using Azure Logic Apps to be delivered to the operations team on time consistently.

Fraud detection model:

- Developed a binary classification model to predict whether a customer is fraudulent or not fraudulent.
- This project lead to discovery of internal agent that were stealing to company
- Created a PowerBI report for the fraud investigation team to track the number of fraud cases, number of accounts investigated, estimated amount lost due to fraud, and a list accounts to be investigated.

Skills

Programming Languages:	Python, SQL, DAX Functions
Machine Learning Frameworks:	Hugging Face Transformers, LangChain, TensorFlow, Pinecone, Natural Language Toolkit (NLTK), Pytorch, Pandas, Pyspark, Scikit-Learn, Darts, Greykite, Decision-Tree, XGBoost, Catboost, Logistic Regression, Pycaret
Natural Language Processing (NLP):	Text Preprocessing, Text Representation, Text classification, Sentiment Analysis, Language Modelling, Named Entity Recognition (NER), Sequence-to-sequence models
AI Chatbot user interface:	Streamlit, Gradio, FlutterFlow, HTML & CSS
Tools & Platforms:	Azure Databricks, Azure Data Factory (ADF), Azure Logic Apps, Azure OpenAI, Azure Storage, Power Automate, Git, Microsoft SQL Management Studio, PowerBI, Visual Code Studio, Copilot, Azure Machine Learning, Azure App Functions, Docker, Modal, Podman, Postman, SQL Server Integration Services (SSIS), PostgreSQL, Microsoft Excel, Azure DevOps
AI Techniques:	Retrieval-Augmented Generation (RAG), Fine-tuning pre-trained models, Semantic Routing, Similarity Search, Re-ranking, Prompt Engineering
Data Engineering	Data Pipelines, Integration Runtime, Extract Transform Load (ETL), Extract Load Transform (ELT), Data Cleaning, Feature Engineering, Embedding

Education

Bachelor of Engineering in Mechanical Engineering (BING Mechanical)
University of Johannesburg

Completed: November 2022

Certifications

Introduction to SQL	Sololearn	2023
SQL intermediate	Sololearn	2023
Programming in Python	Sololearn	2021
Artificial Intelligence in the 41IR	University of Johannesburg	2021
Foundations: Data, Data, Everywhere	Google	2021
Ask Questions to Make Data Driven Decision	Google	2021

References

Available upon request